

MACROECONOMIC PRINCIPLES (MANKIW CHAPTERS 29 & 34)

C211: Global Economics for Managers

Western Governors University

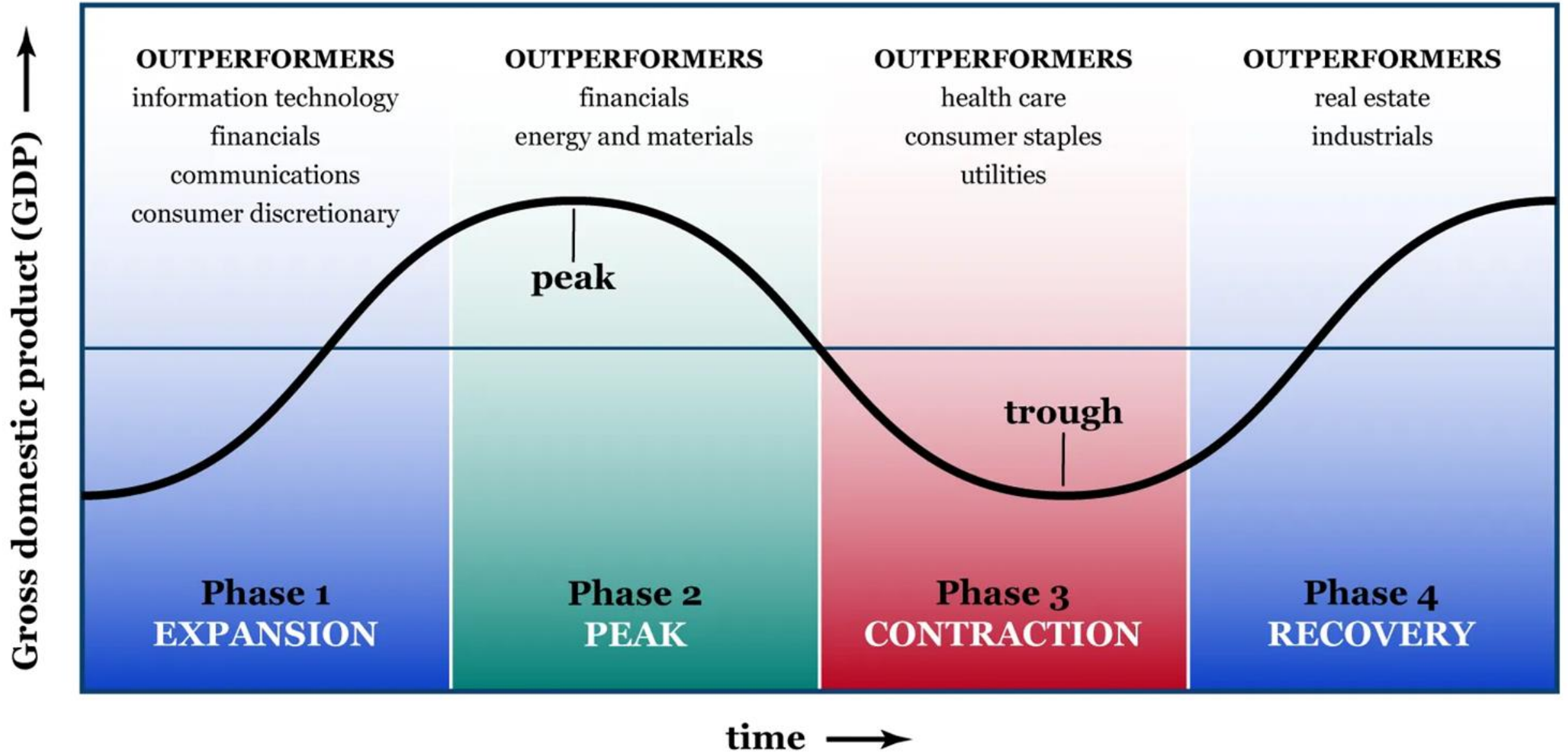


Four phases of an economic cycle

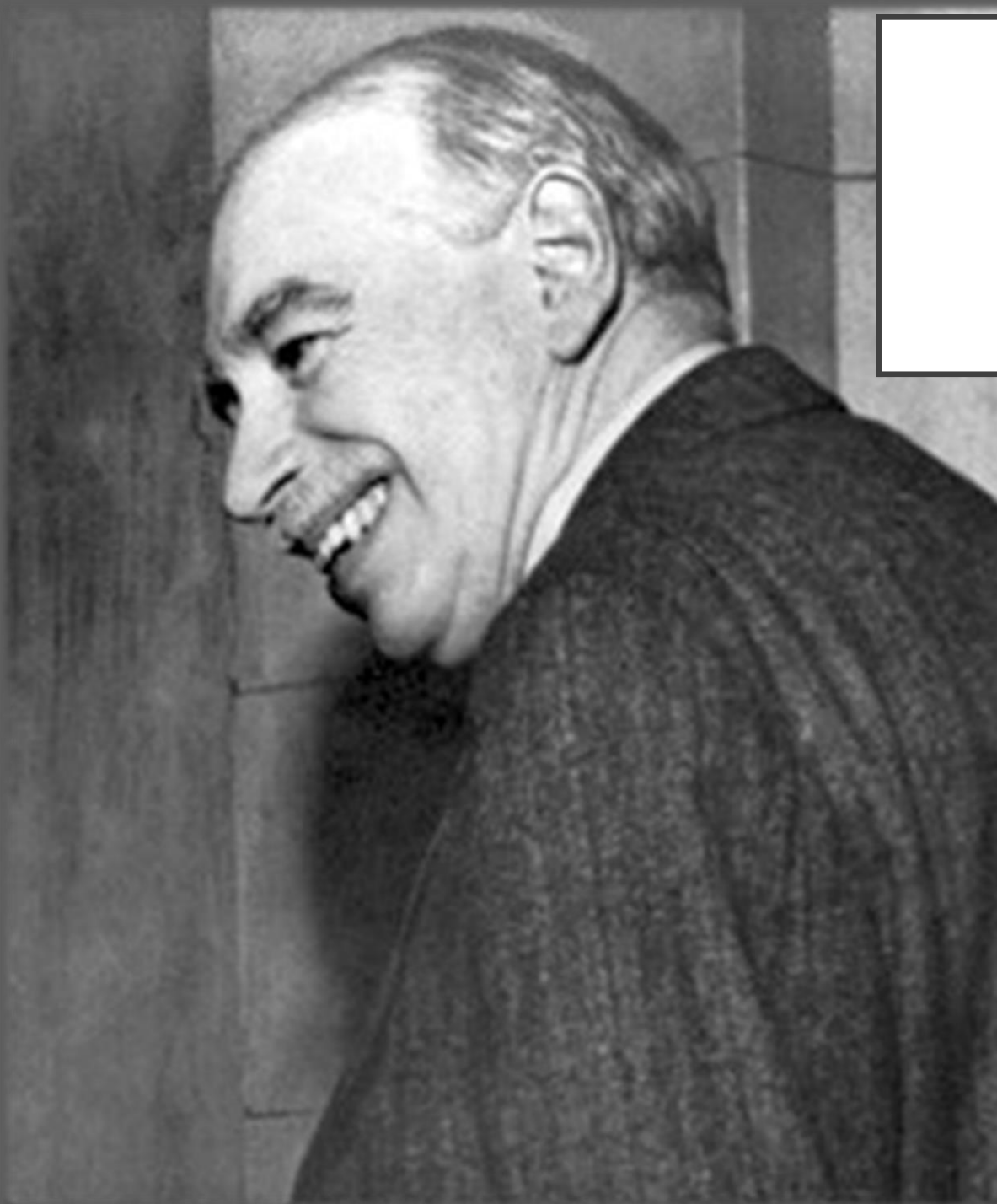
INTRODUCTION

- Originally Microeconomics was all there was
 - Macro was called “Business Cycle Theory”
 - Just took micro and multiplied it by the size of the economy
- Microeconomics always has an equilibrium
 - If price is too high things go unsold and price falls
 - So there is no permanent dis-equilibrium

Four phases of an economic cycle



- Born wealthy
- Studied under Alfred Marshall
 - Wrote the Supply and demand theory
- Bloomsbury Group
- Started his analysis with money



JOHN
MAYNARD
KEYNES

WHAT IS MONEY

- What is Money?
 - Must be 3 things to be money:
 - 1) Store of Value
 - 2) Medium of Exchange
 - 3) Unit of Account



DEFINITION OF MONEY #1: STORE OF VALUE

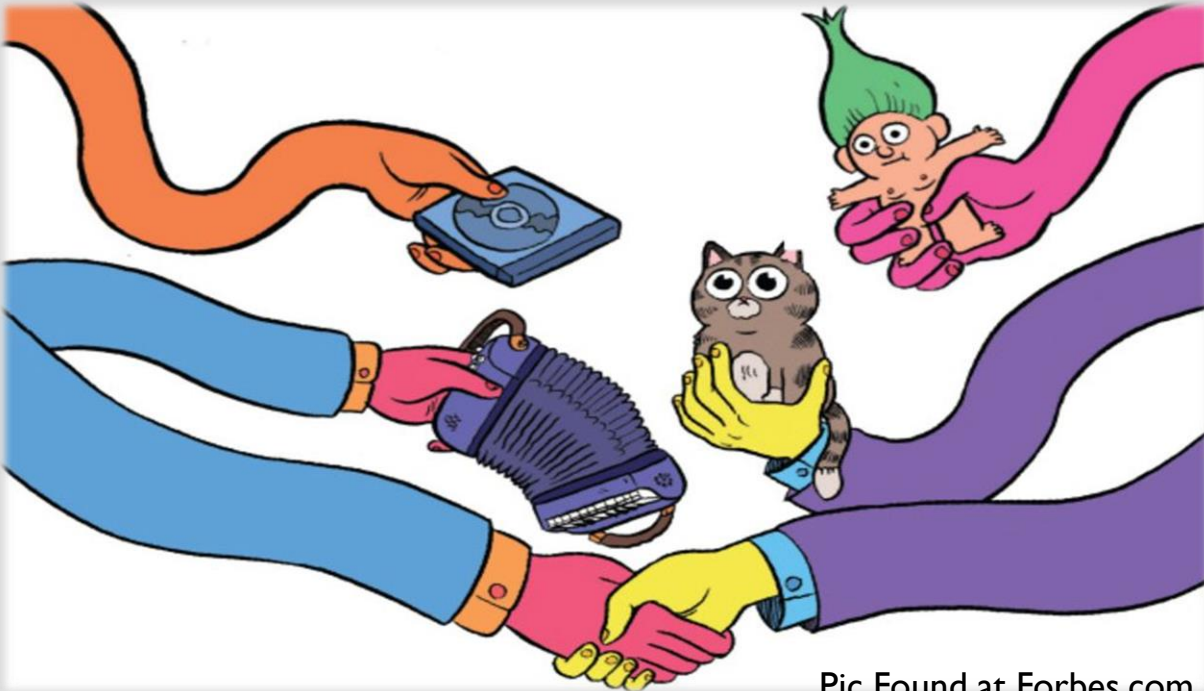
- It must hold its value well
- **Inflation attacks this aspect of money**
- If every time you put a dollar bill in your pocket, it disintegrated that would not work as a currency



Pic found at CBS.com

DEFINITION OF MONEY #2: MEDIUM OF EXCHANGE

- It needs to be accepted in exchange for goods and services or other assets



Pic Found at Forbes.com

- A painting has value, and that value holds over long periods
- But trying to pay for a car with a painting is not going to work
- Money needs to be widely accepted (it needs to be a Medium of Exchange)

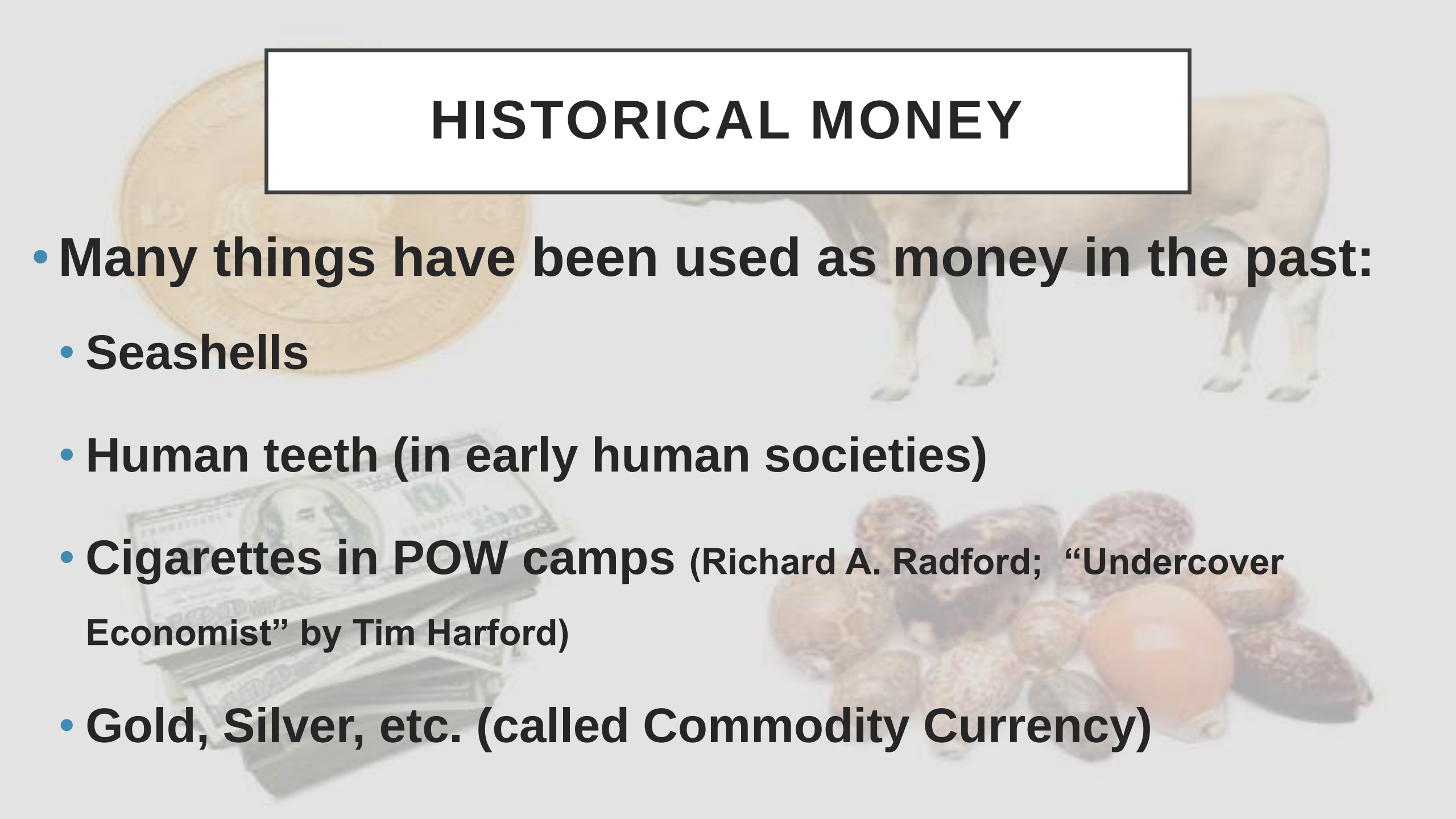
DEFINITION OF MONEY

#3:

UNIT OF ACCOUNT

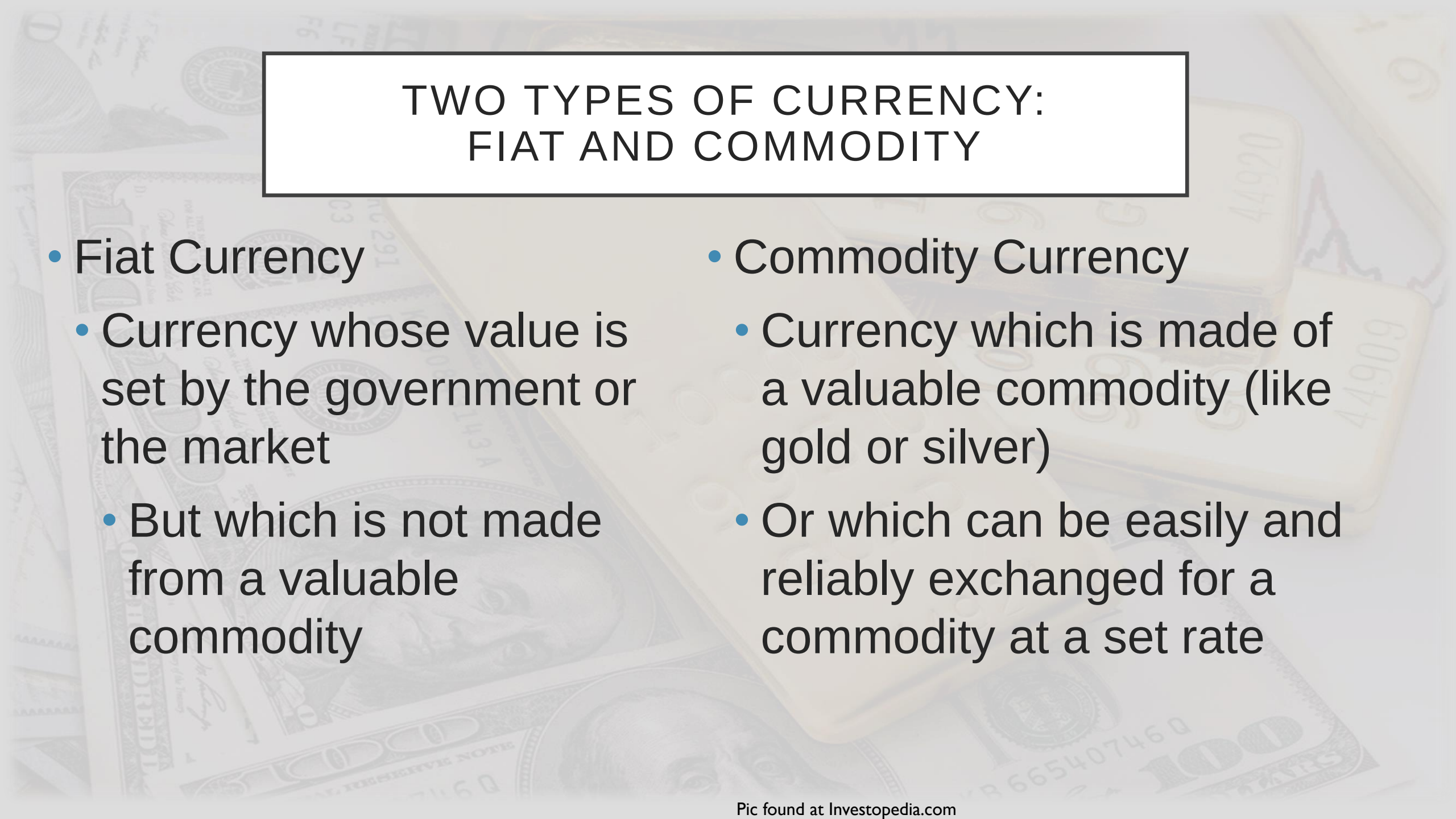
- It needs to be divisible
 - I need to be able to make change and compare different amounts
- Paintings might not be good because what is twice a Van Gogh worth? One-half?
 - Easy to see a \$20 bill is worth twice what a \$10 is





HISTORICAL MONEY

- **Many things have been used as money in the past:**
 - **Seashells**
 - **Human teeth (in early human societies)**
 - **Cigarettes in POW camps (Richard A. Radford; “Undercover Economist” by Tim Harford)**
 - **Gold, Silver, etc. (called Commodity Currency)**



TWO TYPES OF CURRENCY: FIAT AND COMMODITY

- Fiat Currency
 - Currency whose value is set by the government or the market
 - But which is not made from a valuable commodity
- Commodity Currency
 - Currency which is made of a valuable commodity (like gold or silver)
 - Or which can be easily and reliably exchanged for a commodity at a set rate



S15E04



Smiths



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There are eight kinds of paper note above \$100:



Two types of \$500 bill



Two types of \$1000 bill



A \$5000 bill



Two types of \$10,000 bill



\$100,000 gold certificate

It is illegal for a currency note collector to hold a gold certificate.

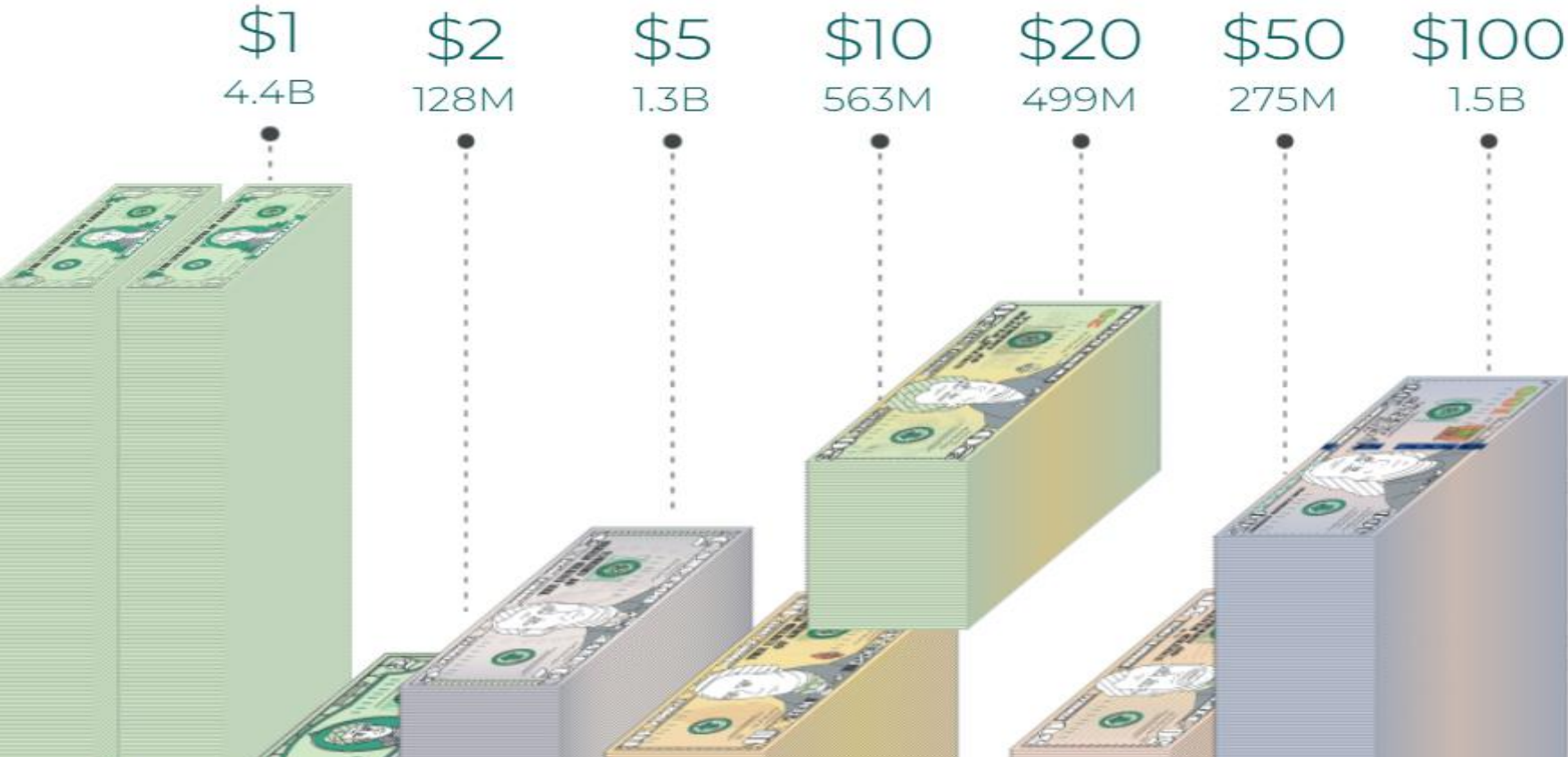


The remaining notes are still legal tender.

Each year, the **FRB** places a print order with the **BEP** to produce new banknotes.

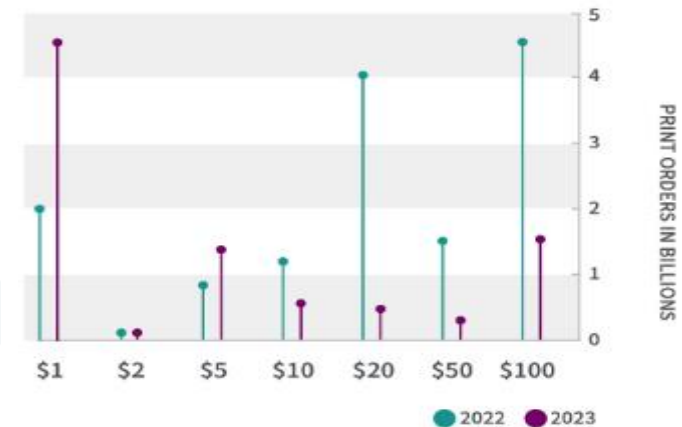
Cash In Circulation

[USCURRENCY.GOV](https://www.uscurrency.gov)



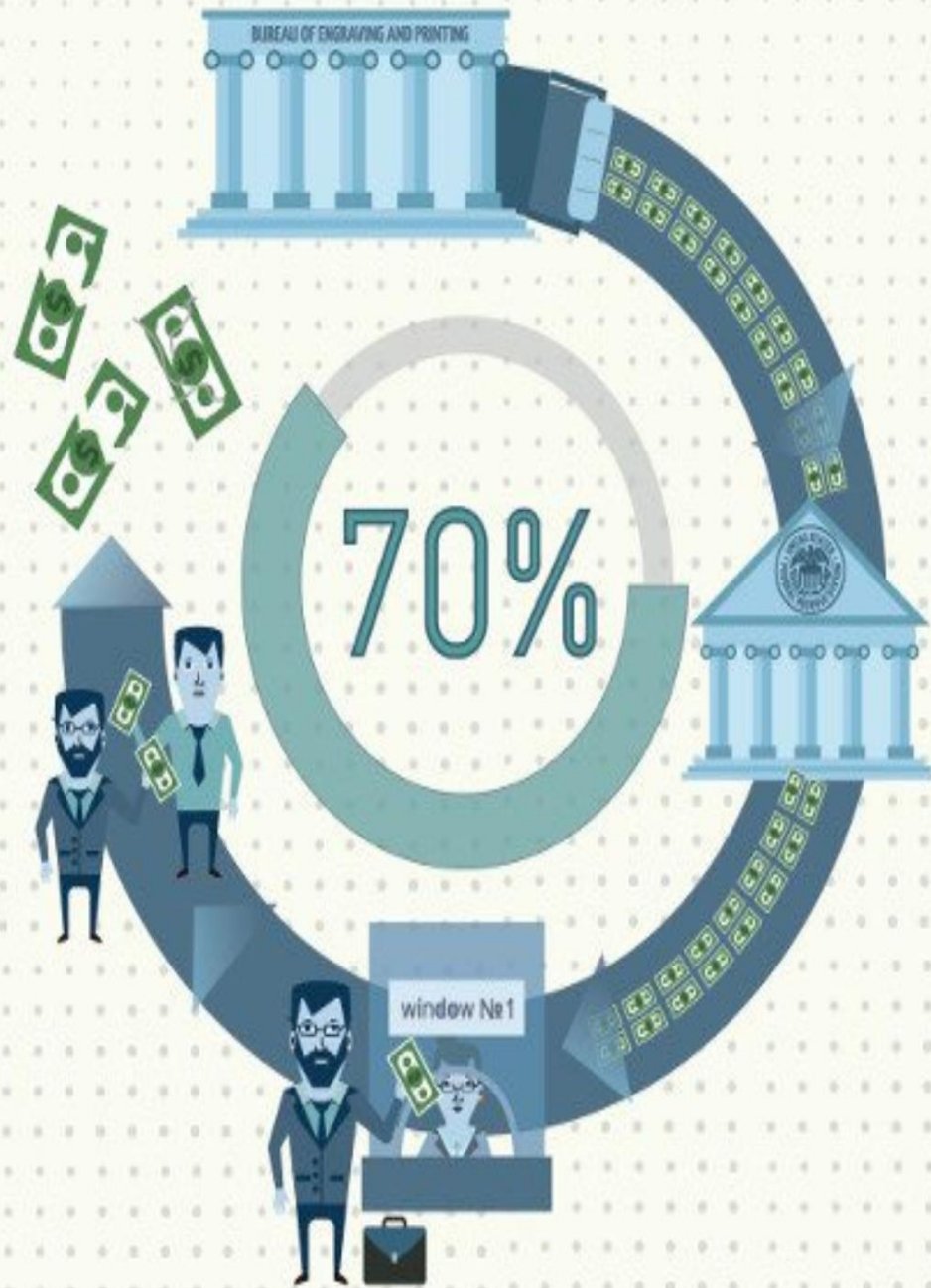
The order is based on the **FRB's** estimate of public demand of currency for the upcoming year and how much currency they estimate will be destroyed because it is unfit to circulate.

2022 VS. 2023 CURRENCY PRINT ORDERS*



There is more than
\$2.2 TRILLION WORTH OF FEDERAL RESERVE NOTES IN CIRCULATION

More than
70%
of the paper notes that
the U.S. Bureau of
Engraving and Printing
delivers each year are
used to replace older
notes going out of
circulation.



HOW DO WE CREATE MONEY?

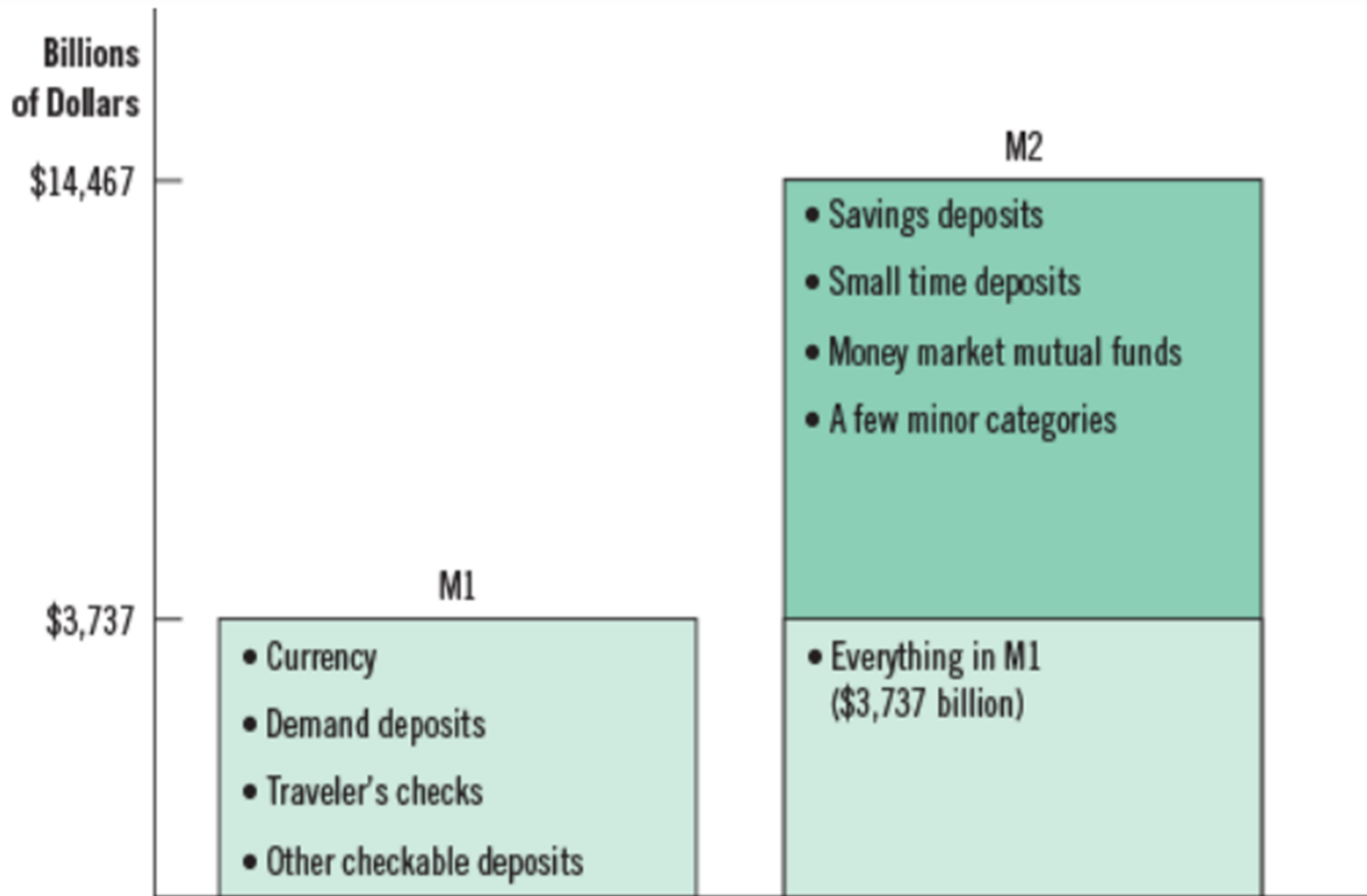
- Creation of paper currency is not the majority of money creation
- Most printed bills and coins are just replacing old money that has become unusable

WHY WRITE
A
CHECK?

USE YOUR
BANK
CARD!



Search "retro renovation"



Source: Federal Reserve.

WHAT IS MONEY: M1 & M2

- Figure 1 ([page 29-1c](#)) in your text shows the total of M1 & M2 in billions of dollars
- Changing the amount of M1 & M2 is “creating money”
- Changing M1 or M2 changes the “Money Supply”

M1 & M2

Credit

Credit Card Balances

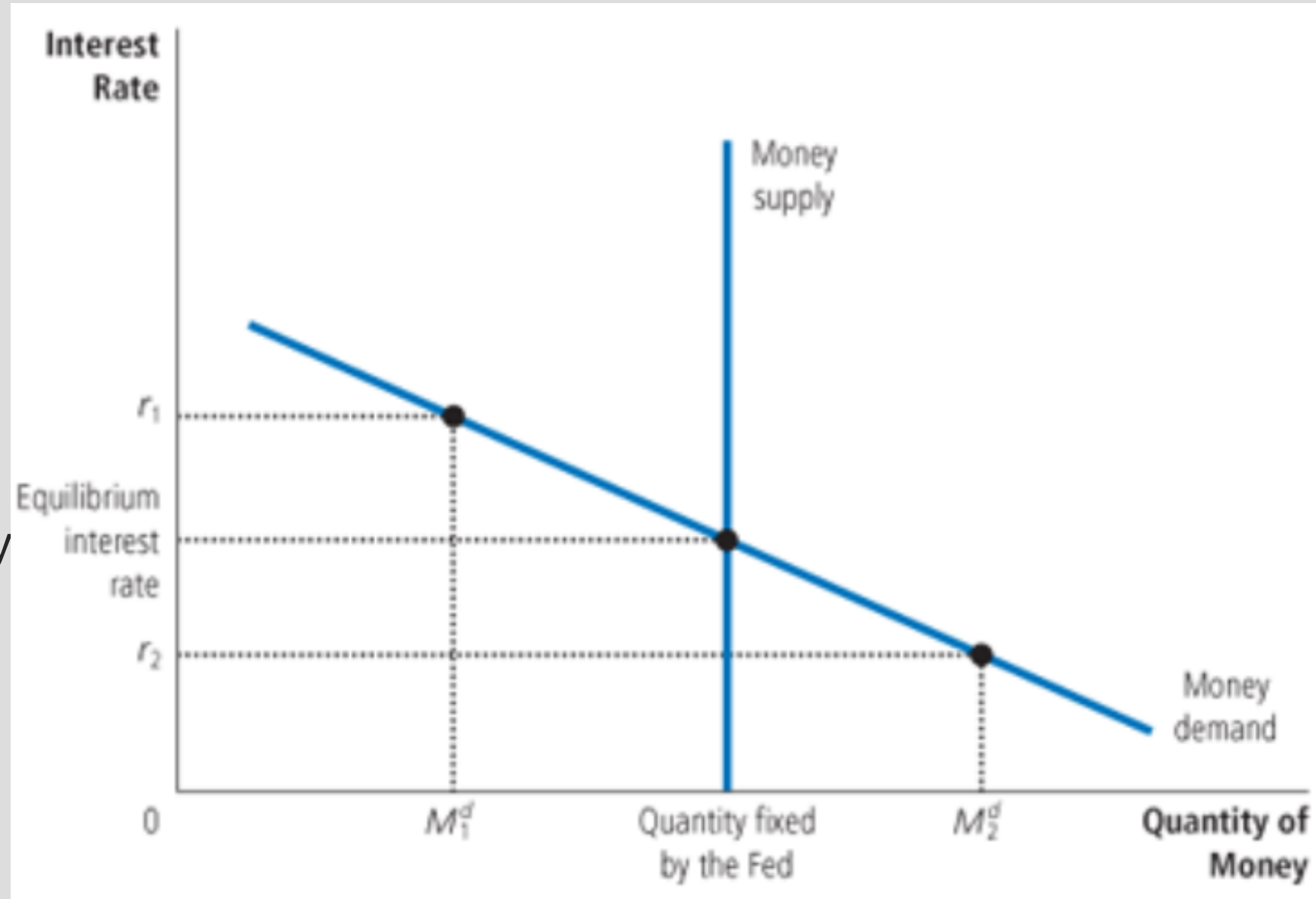
- Decrease one's wealth
- Money is an asset; credit card debt is a liability

Not \$

For this reason, credit cards, even though they are used to purchase items, are never counted as money

MONEY SUPPLY

- The quantity of M2 is the total value of the Money Supply
- When The Federal Reserve sets the quantity of Money Supply
 - That is why the Money Supply is a vertical line



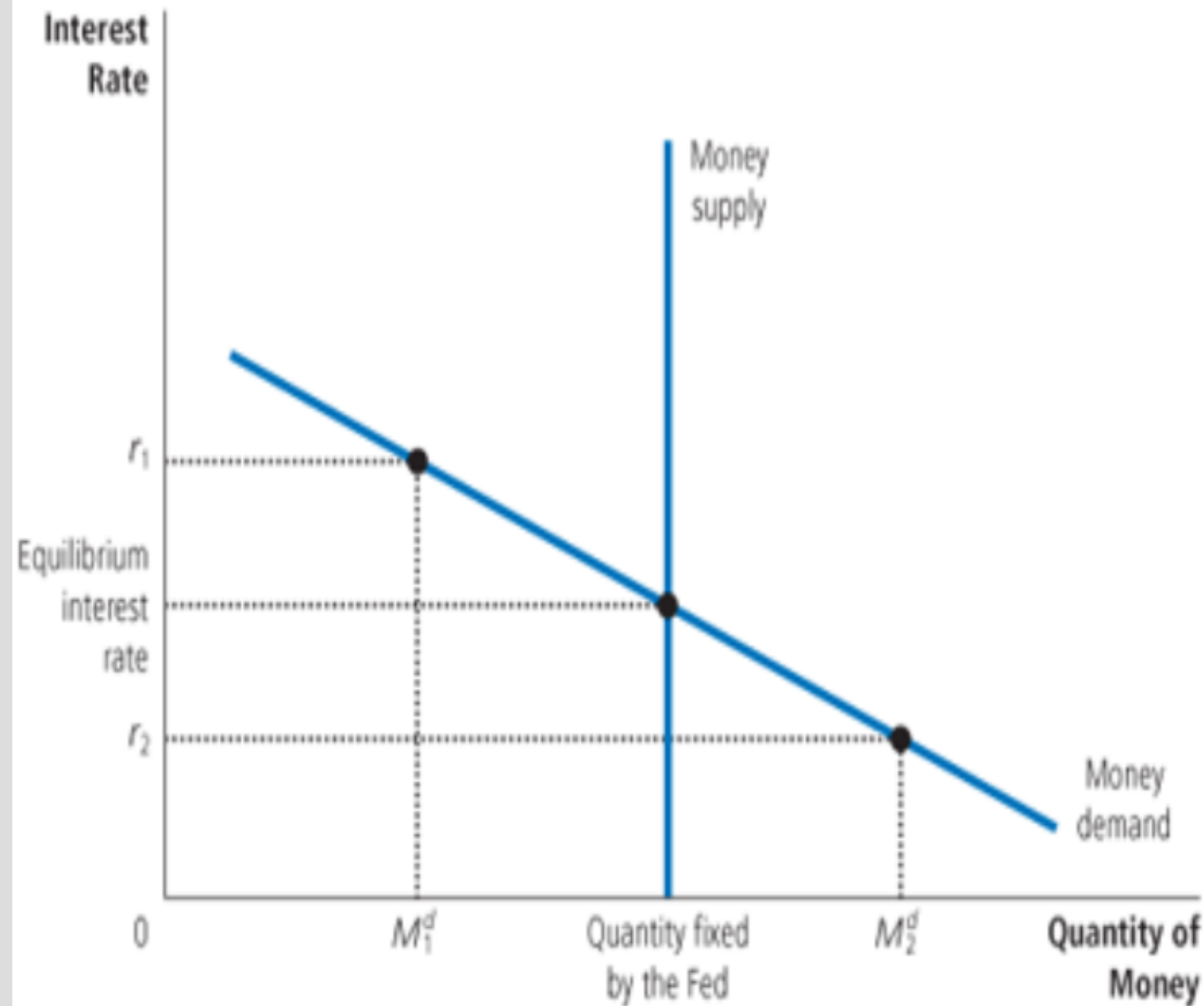
MONEY
DEMAND:
THE THEORY
OF LIQUIDITY
PREFERENCE

- “People choose to hold money instead of other assets that offer higher rates of return because money can be used to buy goods and services.”-([page 34-1a](#))
- People need money for:
 1. Transactions
 2. Emergencies
 3. Speculating on Foreign Exchange Markets

MONEY
DEMAND:
THE THEORY
OF LIQUIDITY
PREFERENCE

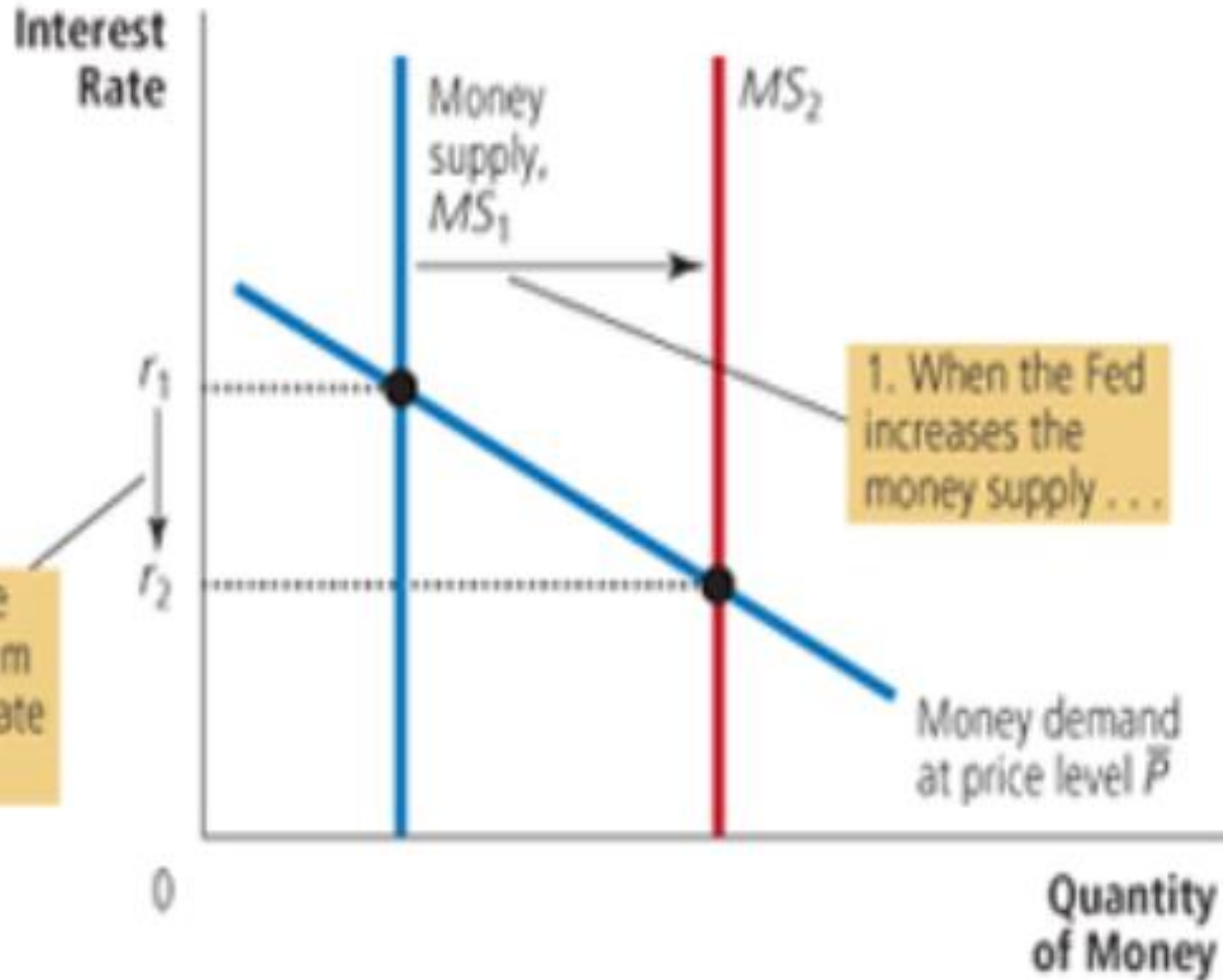
- But if you are holding money (not bonds) then you do not earn any interest on it
- The Opportunity Cost of holding money is the interest rate you could have earned by holding other assets (like bonds)
- If the interest rate is high, people will want to hold less money
 - And if the interest rate is low then you will hold more money

SUPPLY & DEMAND: BUT THIS TIME WITH ~~FEE~~ FEELING MONEY



- The market interest rate is the equilibrium price for money!
 - Also called the Fed Funds Rate
- And when the Money Demand curve Or the Money Supply Curve moves, this will change the Fed Funds Rate (Equilibrium Interest Rate or Market Interest rate)

(a) The Money Market

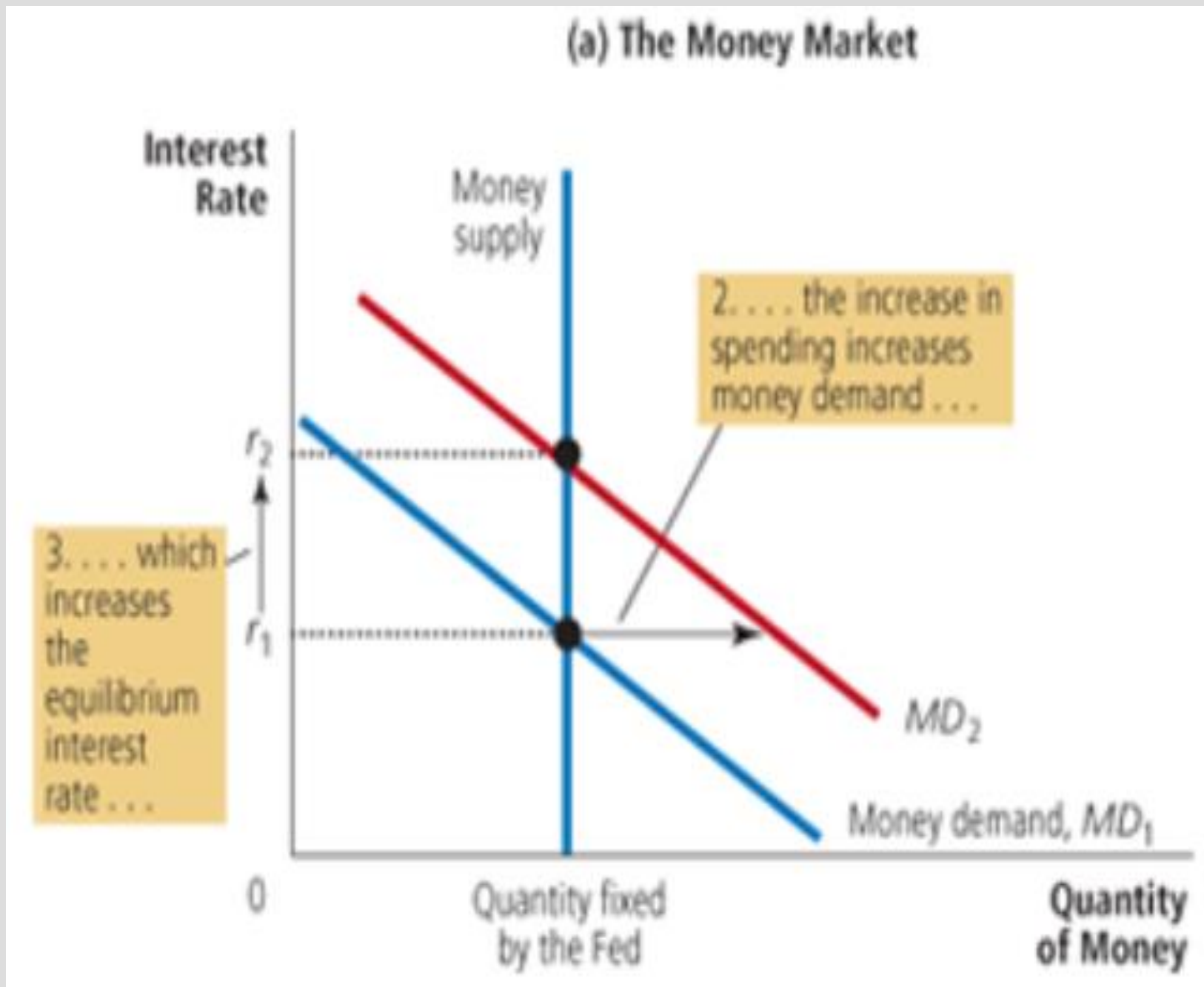


CHANGING THE INTEREST RATE

- When The Fed:
 - Increases the Money Supply
 - The Fed Fund Rate(r) falls
 - Decreases the Money Supply
 - The Fed Funds Rate(r) rises

WORKS JUST LIKE SUPPLY AND DEMAND

- If Money Demand were to rise
 - The Fed Funds Rate(r) rises
- If Money Demand were to fall (shift left)
 - The Fed Funds Rate(r) falls



THE FEDERAL RESERVE SYSTEM



Board of Governors
decides Money Supply



12 Regional banks each
with a Head



1 Chairperson oversees
the Board of Governors

THE FEDERAL
RESERVE SYSTEM:
**MONETARY
POLICY**



Changes in Money
Supply is called
Monetary Policy



It is controlled by The
Federal Reserve



The Federal Reserve
does not have to ask
the president or
congress

TOOLS OF THE FEDERAL RESERVE: MONETARY POLICY



1) Reserve Requirement Ratio

How much the bank must keep as “reserves” at The Federal Reserve



3) Open Market Operations

The buying and selling of US Government Treasuries (bonds)



4) Discount Window Rate

The rate of interest The Federal Reserve charges banks when banks borrow from The Fed

- To meet their reserve requirements
- Usually higher than Fed Funds Rate

2) Interest Paid On Reserves

Just like a savings account
Reserves held at The Fed pays Interest

**Increases the Money Supply
(Decrease the Fed Funds Rate)**

1. Open Market Purchase

-Buy bonds and pays for them with M2 money

2. Decrease the Discount Window Rate

-Cost of borrowing for banks falls so they lend more

3. Decrease Reserve Requirement Ratio

-Banks have more money to lend out
-Increase the Money Multiplier

4. Decrease Interest Paid on Reserves

-With less interest paid on reserves, banks will lend more and save less.

**Decreases the Money Supply
(Increase the Fed Funds Rate)**

1. Open Market Sale

-The Fed takes M2 money in exchange for bonds

2. Increase the Discount Window Rate

-Cost of borrowing for banks rises so they lend less

3. Increase Reserve Requirement Ratio

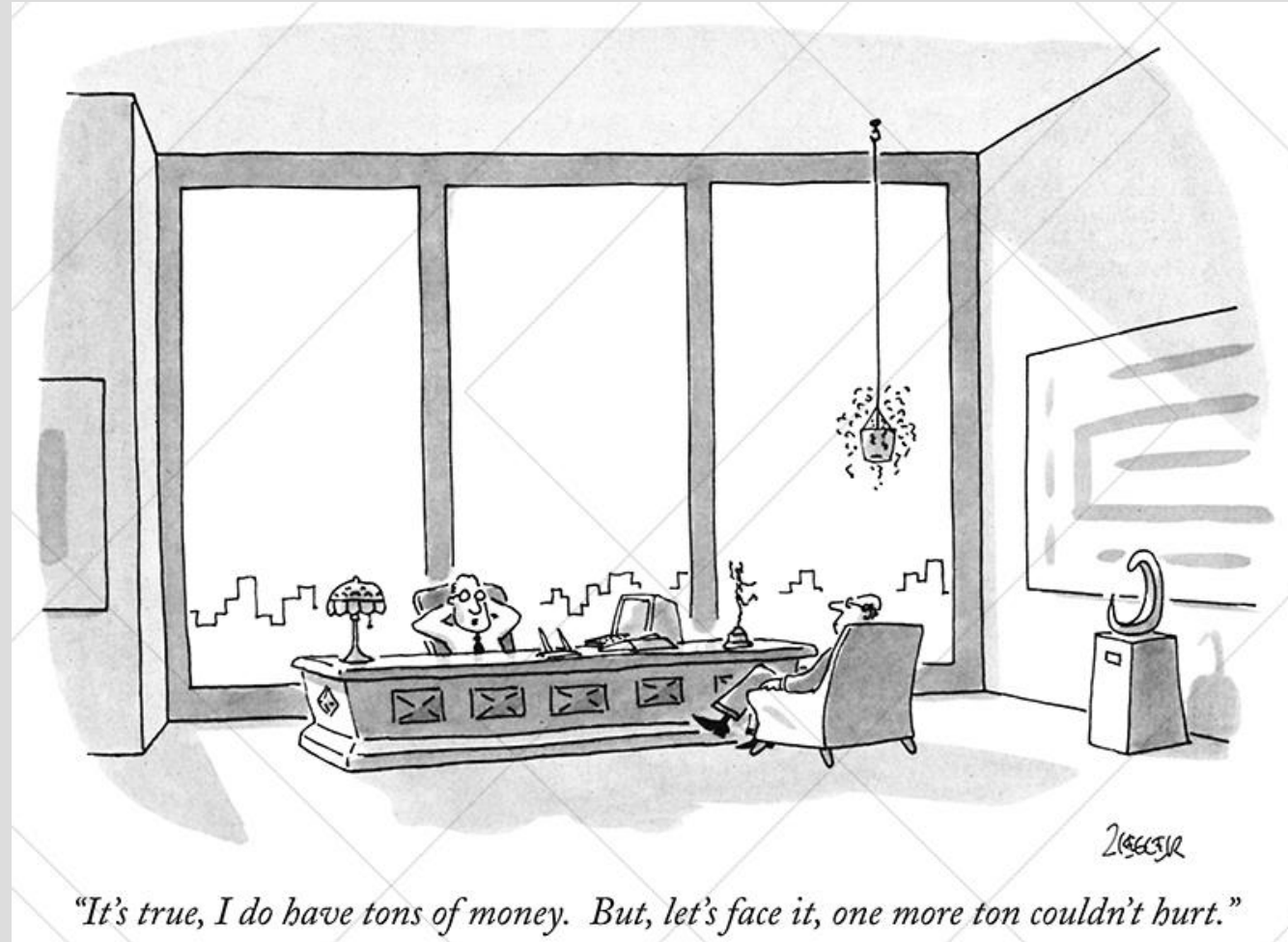
-Banks have less money to lend
-Decrease Money Multiplier

4. Increase Interest Paid on Reserves

-With more interest paid on reserves, banks will lend less and save more.

BUT HOW IS MONEY CREATED?

- Via bank loans!
- Fractional Reserve banking
 - The banking system is only required to hold onto a portion of deposits
 - They loan out the rest creating money
- The Federal Reserve changes how much money banks lend, by using the Tools of Monetary Policy



FRACTIONAL RESERVE BANKING

If the Reserve Requirement is 100% then they would have nothing left over to loan out!

Assets		Liabilities	
Reserves	\$100.00	Deposits	\$100.00

- If the Reserve Requirement is less than 100%, they have money to loan out
- Here is an example with a 10% Reserve Requirement

Assets		Liabilities	
Reserves	\$10.00	Deposits	\$100.00
Loans	\$90.00		

BANK LOANS AND MONEY SUPPLY

1. Someone deposits \$100 and the Reserve Requirement is 10%

Assets		Liabilities	
Reserves	\$10.00	Deposits	\$100.00
Loans	\$90.00		

2. The loan of \$90 becomes a deposit at the next bank who can lend out \$81 (\$90 minus 10%)

Assets		Liabilities	
Reserves	\$ 9.00	Deposits	\$90.00
Loans	81.00		

3. That \$81 loan becomes a deposit at the bank of whoever took out that loan

- And that bank can now lend out \$72.90 (\$81 minus 10%)

Assets		Liabilities	
Reserves	\$ 8.10	Deposits	\$81.00
Loans	\$72.90		

BANK LOANS AND MONEY SUPPLY

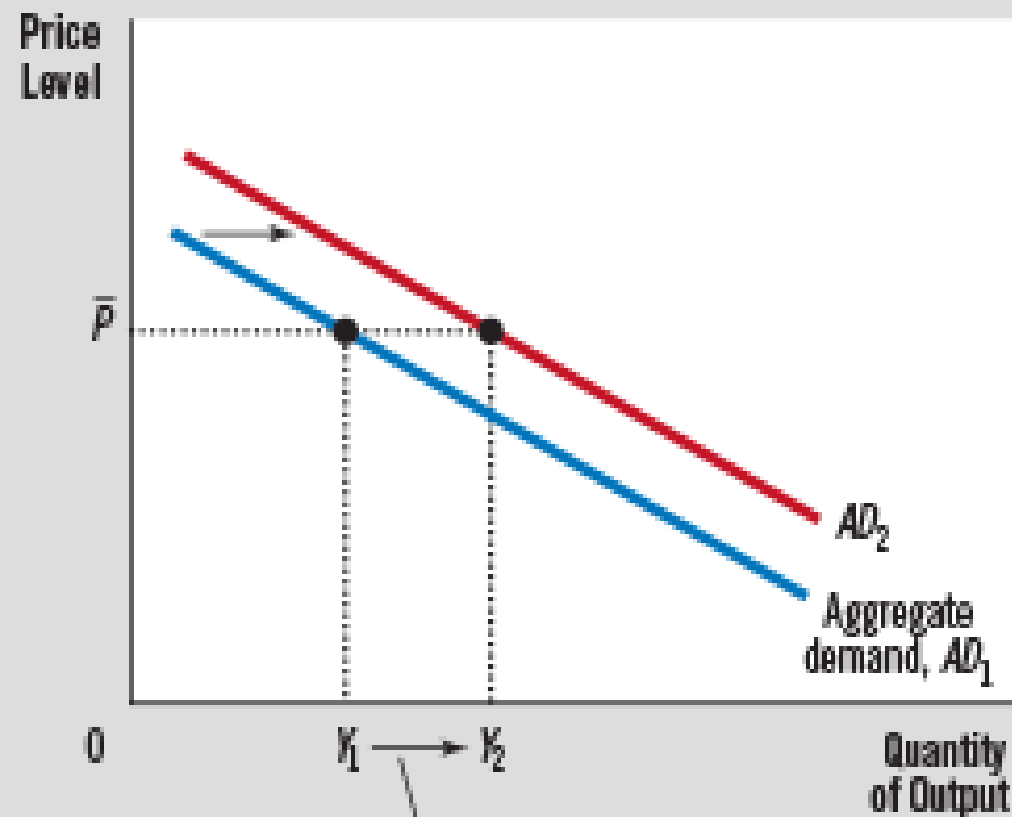
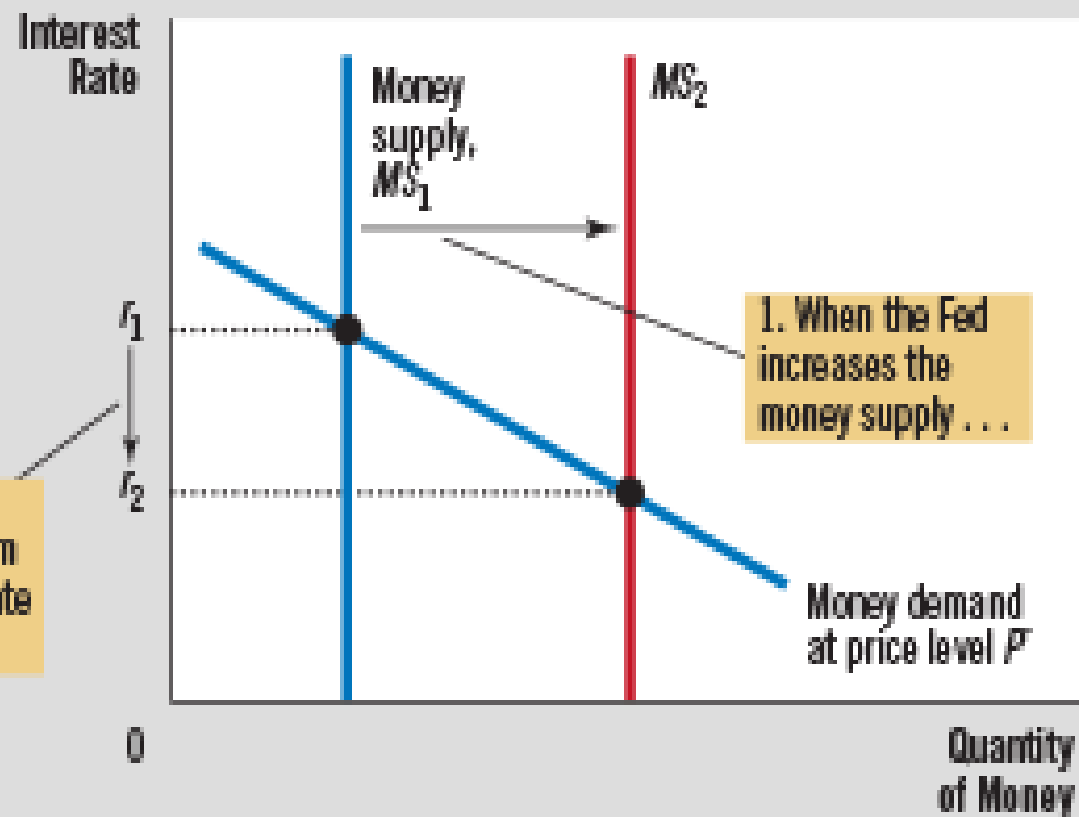
Original deposit	= \$100.00
First National lending	= \$ 90.00 (= .9 × \$100.00)
Second National lending	= \$ 81.00 (= .9 × \$90.00)
Third National lending	= \$ 72.90 (= .9 × \$81.00)
•	•
•	•
•	•
Total money supply	= \$1,000.00

- This process of loan → deposit → loan creates money
- Each loan increases the Money Supply
- In our example we started with a Money Supply = \$100
- And we end with a Money Supply = \$1,000

AN INCREASE IN MONEY SUPPLY

(a) The Money Market

(b) The Aggregate-Demand Curve



BANK LOANS AND MONEY SUPPLY

What else can stop this process?

- If banks do not want to lend
- If people do not want to borrow

Both of these happened after 2008

FINAL STAGE: THE MONEY MULTIPLIER

- The book explains it like this:

- $$\text{Money Multiplier} = \frac{1}{\text{Required Reserve Ratio}}$$

- $$\text{Total Deposits @ End of Process} = \text{Money Multiplier} * \text{Initial Deposit}$$

- In our Example:

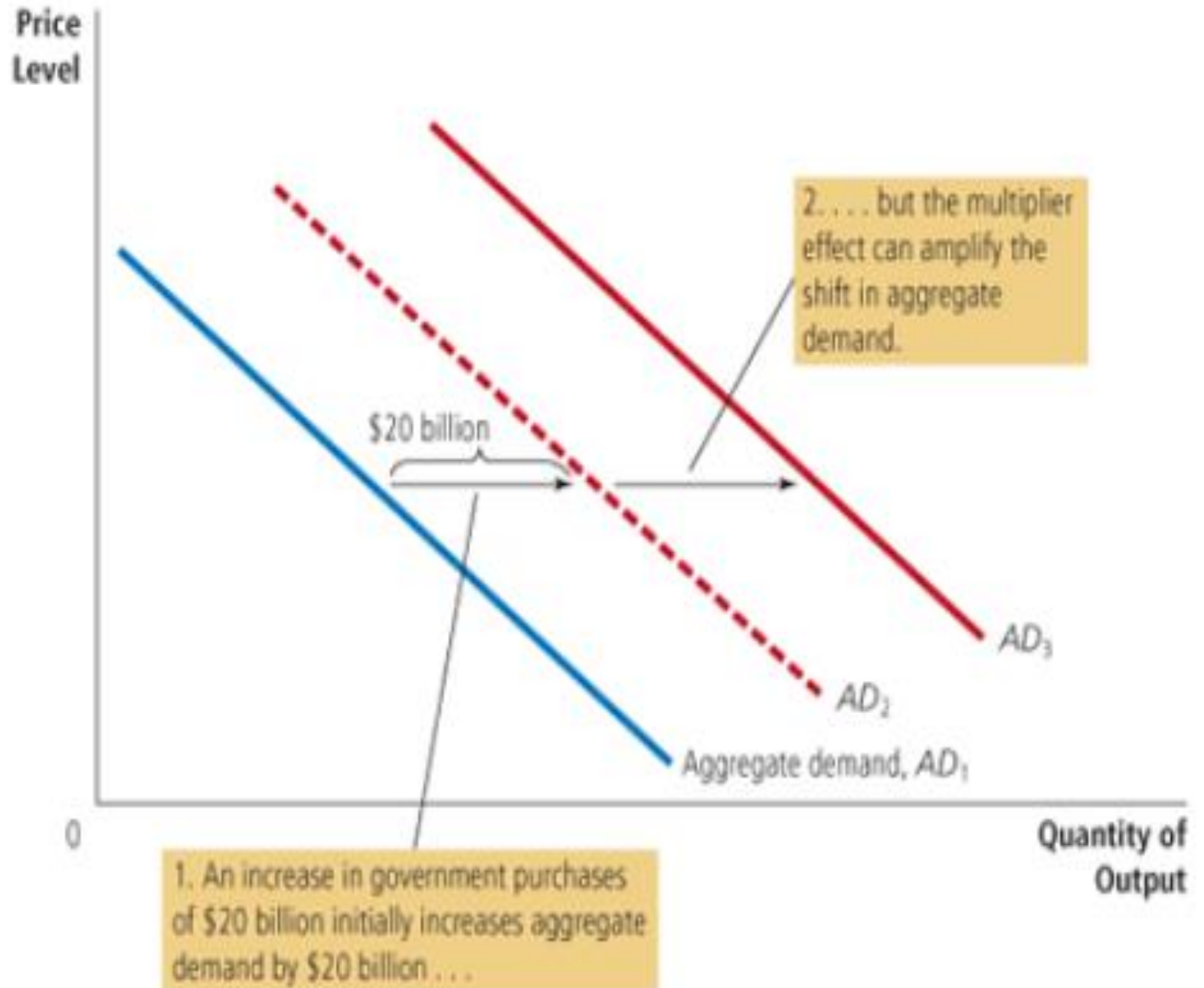
- $$\text{Total Deposits @ End of Process} = \frac{1}{10\%} * \$100 = \$1,000$$

CHANGING
AGGREGATE
DEMAND

- Fiscal Policy
 - Controlled by Congress
 - Controls Government Expenditure and Taxes
- Monetary Policy
 - Controlled by The Federal Reserve
 - Controls the amount of money in the economy and thus the market interest rate

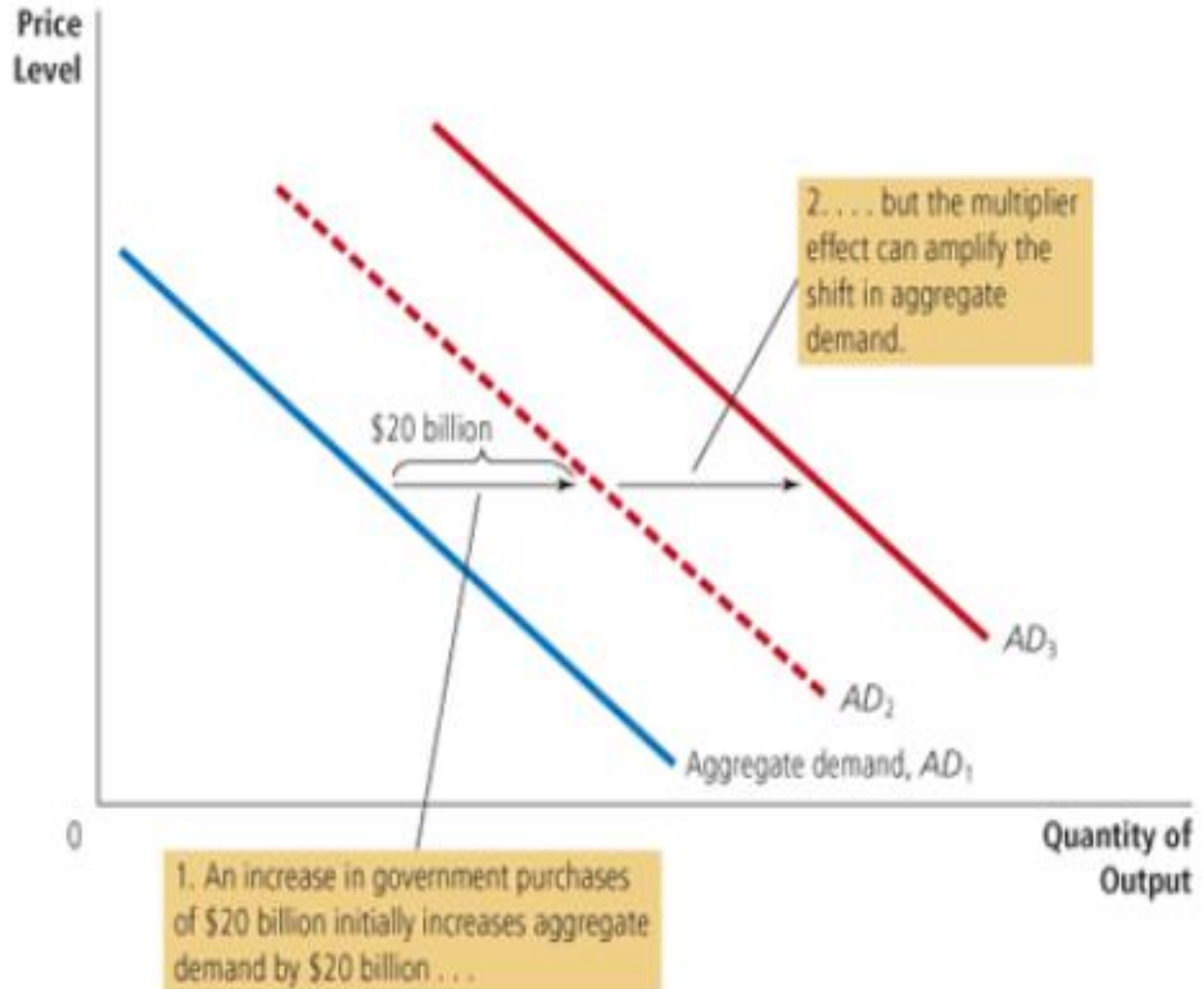
FISCAL POLICY

- Fiscal Policy
 1. Government Expenditure
 2. Taxes
- Increases in Government Expenditure, or increases in Consumption (because people pay less in taxes) cause an increase in Aggregate Demand
- This is shown by an increase of \$20 *billion* (from the blue line to the dotted red line)



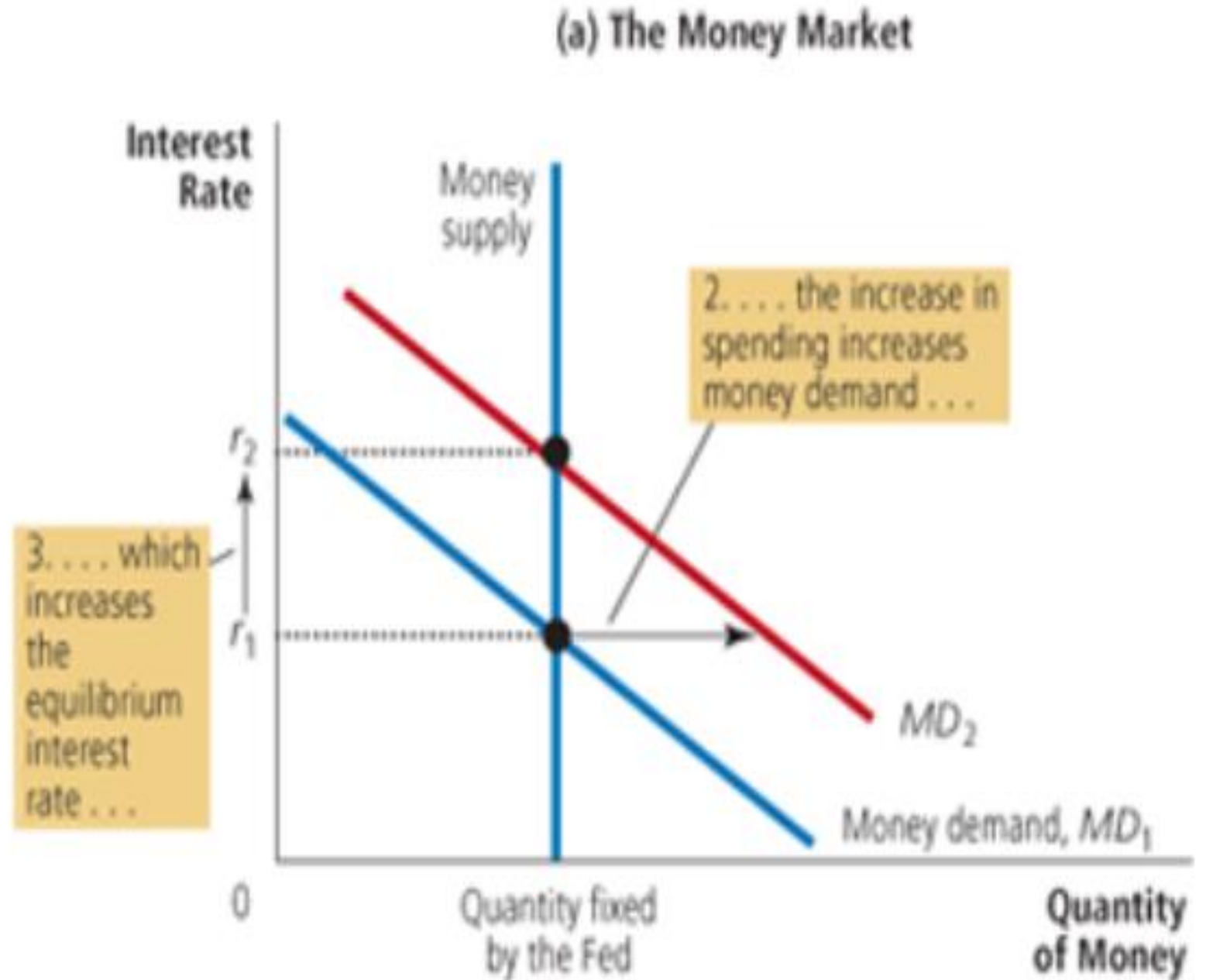
EXPENDITURE MULTIPLIER EFFECT

- Every person's spending is another person's income
- The initial increase in spending becomes income
 - And that income for the second person becomes new spending
 - Becomes income for a 3rd person...
- Aggregate Demand moves from the dotted red line to the solid red line



CROWDING OUT EFFECT

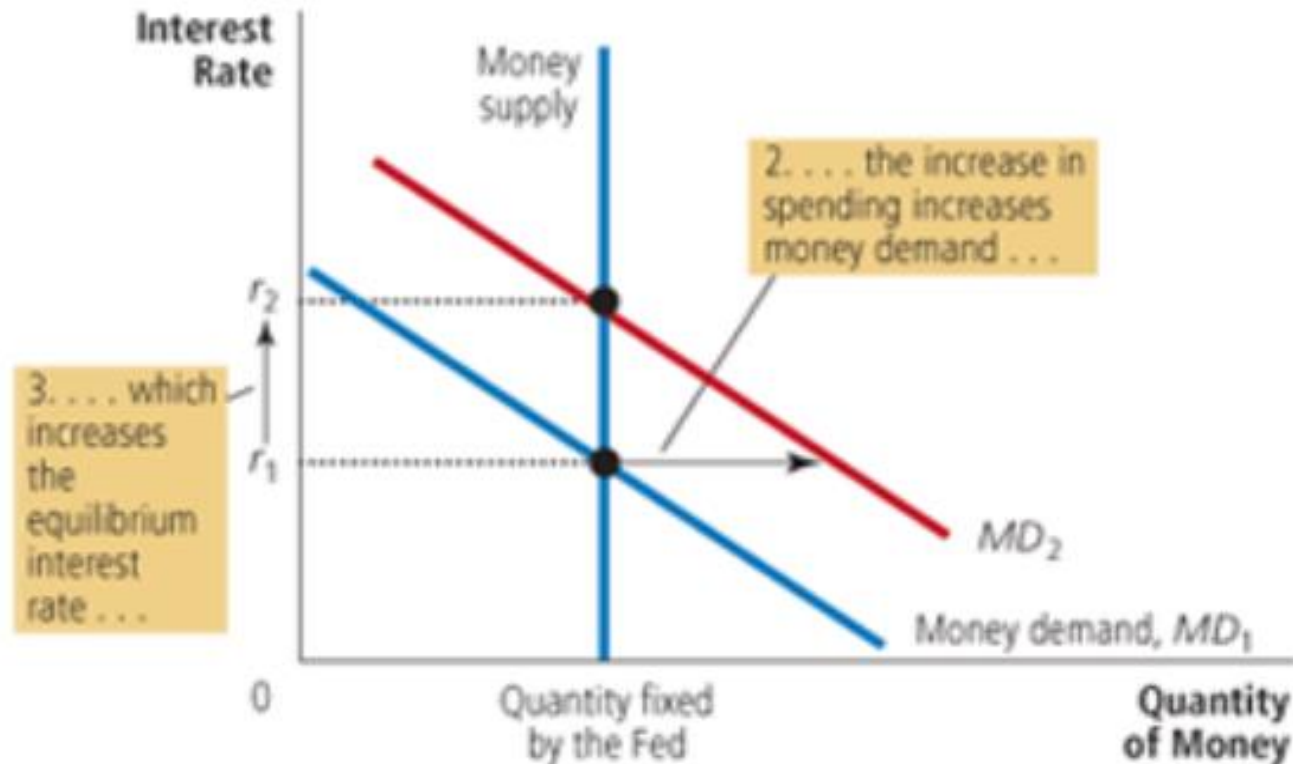
- Fiscal Policy has a drawback
- Increases in Government Expenditure, or reduced taxes can cause an increase in Money Demand
- Money Demand increases and this increases the market interest rate (r)



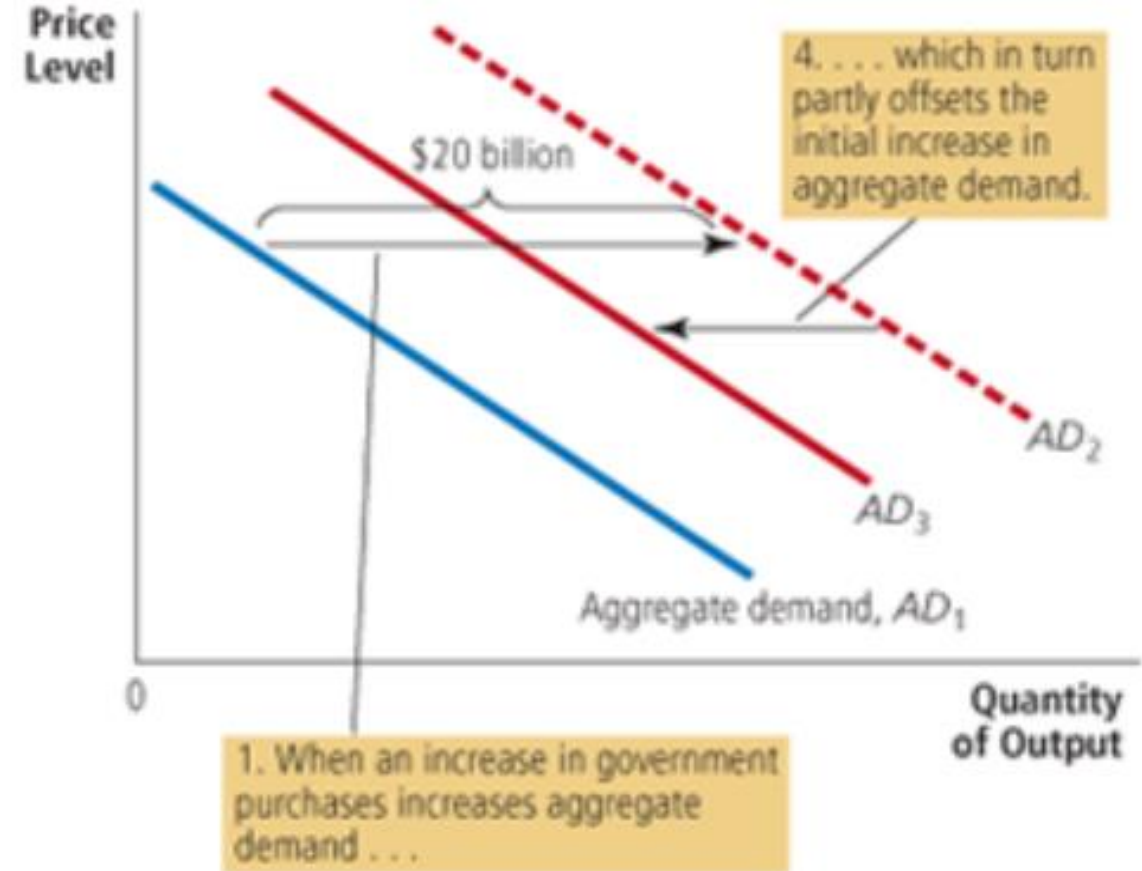
CROWDING OUT EFFECT

- Expansionary Fiscal Policy increases Aggregate Demand
- This pushes up Money Demand with rising incomes
- Causing the Fed Funds Rate to increase
- Causing private investment to fall

(a) The Money Market



(b) The Shift in Aggregate Demand



CONCLUSION

Monetary Policy is The Tools of The Federal Reserve

- Changing the Money Supply in order to change the Fed Funds Rate

Fiscal Policy is Government Expenditure and Taxes

- Changes the level of Government Expenditure and Consumption (via taxes)
- Possible Crowding Out Effect